Air Quality Forecasting on the UCI Dataset: Preprocessing, Regression, and Classification

COMP9417 Group Project
Group-GoodJob

1 Introduction

This report investigates forecasting tasks on the UCI Air Quality dataset, addressing both continuous concentration prediction (regression) and risk stratification (classification). Models are trained on year 2004 data and evaluated on year 2005 data to emulate real-world deployment with strict chronological separation. Performance is reported using standard metrics (RMSE, MAE, and R^2 for regression; accuracy and confusion matrices for classification). The project emphasises a fully reproducible preprocessing pipeline that produces ready-to-use feature packs for different model families.

2 Data Analysis

The original CSV provides timestamped measurements for multiple pollutants and environmental variables with European decimal formatting, occasional gaps, and sentinel values. Preprocessing converts this into a cleaned, gap-aware, time-aligned table, adds lags and rolling statistics, derives target variables for both tasks, and writes out feature packs. Deterministic splits keep all of 2004 for train/validation and all of 2005 for test, avoiding temporal leakage and clarifying generalisation behaviour.

3 Methodology

Preprocessing and Feature Engineering

Goal. Turn the raw UCI CSV into a cleaned, gap-aware, time-aligned table and two ready-to-use feature packs: (i) a raw-scale pack for tree models and (ii) a standardised pack for neural models. Steps are scripted for full reproducibility.

Pipeline summary.

1. **Read & parse.** Load the semicolon-separated CSV with European decimals; combine `Date` and `Time` into `timestamp` and sort chronologically.
2. **Clean missing/sentinel values.** Replace known sentinels (e.g., `-200`) with `NaN`. Forward-fill short gaps within a small window; retain a binary `gap_flag`.
3. **Targets.** Create next-horizon regression targets for CO and thresholds for classification using fixed cutoffs (1.5 and 2.5 mg/m^3) per project requirement.
4. **Features.** For each sensor series: create lags (e.g., 1, 2, 3, 4), rolling means and standard deviations with windows $W \in \{3, 6, 12\}$ (with `min_periods` to avoid dropping many rows). Add time features (hour, day-of-week, month, weekend) and retain the `gap_flag`.
5. **Packs.**
   (a) `trees/`: raw-scale features for tree/boosting models.
   (b) `nn/`: same columns but standardised with `StandardScaler`.
6. **Splits.** Deterministic chronological split: all of 2004 used for training/validation; all of 2005 used for testing. A JSON file stores train/val/test indices.

Regression methodology

In this regression model development, we use the data packs prepared in the preprocessing stage to forecast the concentrations of four pollutants in the air-quality dataset — CO(GT), C6H6(GT), NOx(GT) and NO2(GT) — at horizons of 1, 6, 12 and 24 hours into the future. All models were trained using 2004 data and evaluated using 2005 data (in chronological order) to simulate real-world deployment scenarios. All the models are compared against the native persistence baseline using RMSE (primary), MAE, and R^2. Linear regression models (OLS, Ridge, Lasso, Elastic Net), tree-based models (Decision Tree, Random Forest, ExtraTrees, Gradient Boosting, XGBoost), neural network models,

and margin-based models (MLPRegressor and LinearSVR) are evaluated and compared with the baseline. Hyperparameters for the linear and tree-based models are tuned with GridSearchCV, while the neural network models are tuned with RandomizedSearchCV, with training parallelised across 208 CPU cores.

We built an hour-of-day mean baseline prediction for the test set using MAD-z. Points with zt¿3.5 are flagged as anomalies, We output three plot types: (1) time-series view (2) anomaly rate by hour-of-day (3) precision–recall curves against a high-value / sudden-jump quantile proxy. Results show anomalies concentrate in 6–12 h and night/early morning, which is consistent with the model performance dip at mid-range horizons.

Classification methodology

Model Approach: Gradient Boosting is applied for classification on UCI Air Quality dataset, aiming to categorise air quality levels into three classes (0: Safe, 1: Moderate, 2: Hazard) based on pollutant concentrations. The model uses engineered time-series features (lagged values, rolling statistics, time-of-day indicators) derived from the data packs generated in preprocessing. Class boundaries for CO were set using fixed thresholds in line with project requirements. The training uses 2004, and testing uses 2005.

4 Results

Regression results

Output.

model RMSE MAE R2

ElasticNet 46.672014 33.936031 0.423318

Lasso 46.532296 33.962008 0.420114

Ridge 46.577342 34.134779 0.395683

XGBoost 50.730647 36.344354 0.392727

LinearSVR 46.620254 34.185430 0.387722

Random Forest 50.949566 36.404905 0.348016

LinearRegression 53.001792 38.636786 0.334828

MLP 61.263338 44.941325 -0.058078

Naive Baseline 63.92049 48.340017 -0.139530

Table 1: Average model performance (across all pollutants and horizons)

This reports the average performance of all regression models across all pollutants and forecast horizons. Overall, the best results are obtained by Elastic Net and Lasso, indicating that a sparsely-regularised linear relationship captures most of the predictable variation in the data. Next are the various tree-based models and the neural network model, but all of which perform better than the native baseline.

pollutant	RMSE	MAE	R2
CO(GT)	0.989978	0.760195	0.370843
C6H6(GT)	4.870377	3.697864	0.266196
NO2(GT)	37.925455	29.146488	0.417041
NOx(GT)	139.432868	101.879438	0.350813

Table 2: Analysis of the difficulty of pollutant prediction

Table 2 summarises the average prediction difficulty for each pollutant. In relative terms, NO2(GT) is the easiest to predict, with the highest (0.42), followed by CO(GT) and NOx(GT) which have moderate around 0.35–0.37. C6H6(GT) is clearly the hardest pollutant to model, with the lowest .

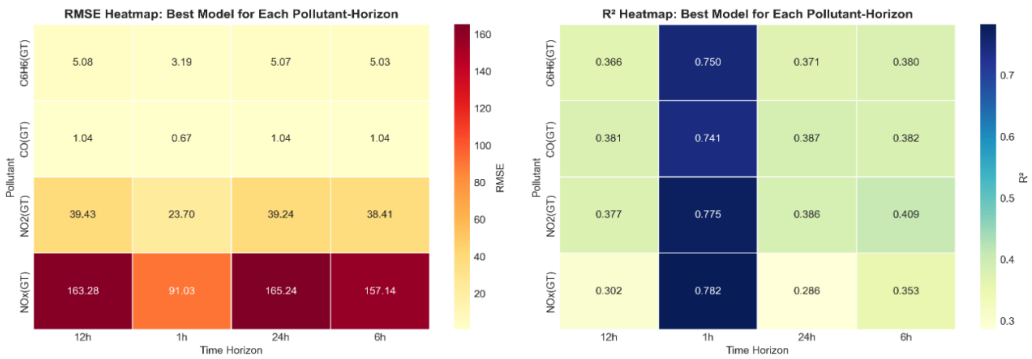

Figure 1: RMSE and R2 heatmap: Best model for each Pollutant-Horizon

Figure 1 shows a heatmap summary of the optimal model for each pollutant-prediction timeframe combination. The left panel reports the root mean square error (RMSE), and the right panel reports R^2.

For all four pollutants, the 1-hour prediction timeframe has the lowest RMSE and the highest R^2 (approximately 0.74–0.78), indicating that short-term predictions are relatively easy once recent historical data is incorporated. As the prediction timeframe increases to 6 hours, 12 hours, and 24 hours, the error increases, and R^2 decreases to approximately 0.28–0.41, indicating that the longer the prediction timeframe, the worse the predictive ability. RMSE values also highlight the different physical scales of the pollutants: CO(GT) and C6H6(GT) have very small absolute errors, whereas NO2(GT) and especially NOx(GT) have much larger RMSE in their own units. The heatmap, which is scale-free, confirms that all pollutants remain moderately predictable at longer horizons, but with noticeably weaker fit compared to the 1-hour case.

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Classification result

Evaluation and Confusion Matrices: Performance is summarised by accuracy and class-wise errors. Confusion matrices indicate that the model distinguishes Safe vs. non-Safe reasonably well, while misclassifications between Moderate and Hazard are more frequent due to overlapping signal and class imbalance.

Key Findings: Gradient Boosting achieves stable overall accuracy with better recall for the majority (Safe) class. Feature importance suggests recent lags and short-window rolling statistics contribute most to discrimination. Further gains would likely require additional exogenous variables and cost-sensitive calibration.

5 Discussion

5.1 Result

Among the various models, regularized linear models (Ridge, Lasso, ElasticNet) performed best overall, indicating that for this type of time series regression task, linear models combined with appropriate regularization can achieve good results. Tree-based models performed relatively poorly due to the limited number of samples and the limited number of truly strong nonlinear structures in the data. Models like MLPs, on the other hand, had very large data sets, leading to insufficient data and making them prone to overfitting the training set, resulting in low R^2 values.

5.2 Limitations and improvement methods

Limited data scale and exogenous variables have constrained the advantages of many models. Future development could incorporate "future-aware" data such as weather forecasts, traffic events, and emission inventories. Alternatively, exploring more time-series features, such as Fourier transform features, may prove beneficial.

Strict temporal segmentation (using 2004 for training/validation and 2005 for testing) prevents data leakage, but it also means that distribution shifts or seasonal mismatches have a more pronounced impact on the model. Exploring the integration of multiple models could potentially improve prediction performance.

6 Conclusion

The pipeline delivers reproducible data preparation and feature packs, enabling side-by-side evaluation of a broad family of models. On strict chronological splits (2004→2005), Elastic Net/Lasso provide the strongest regression performance; Gradient Boosting yields reliable multi-class classification with most errors on adjacent risk levels. The assets (`parquet` features, `splits.json`, and loading helper) support further experimentation without revisiting preprocessing.

7 References

References

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[2] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.